

Co2:

space, subspace, span

homogeneous eqn; linear independence, dependence.

row, left, column, right & null spaces, rank.

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Vector space: vector space consists of a set of vectors and set of scalars.

"Closed" under vector addition and scalar multiplication. form a vector space.

(a) closure prop: for $u, v \in V$, $u+v \in V$

(b) Associative: $u, v, w \in V$, $(u+v)+w = u+(v+w)$

(c) Existence of additive identity, $\rightarrow$ zero element;

$\exists e \in V$ such that any $u \in V$

$$u+e = e+u = u$$

$e \rightarrow$ zero element.

(d) existence of additive inverse;

for any $u \in V \exists v \in V$ such that

$$u+v = v+u = e$$

(e) commutative; for any $u, v \in V \exists u+v = v+u$.

=

(1) for any $c \in F$, $u \in V \exists c*u \in V$

(2) $c_1, c_2 \in F$, $u \in V \Rightarrow (c_1+c_2)u = c_1*u + c_2*u$

(3) $1 \in F$, $u \in V \Rightarrow 1*u = u*1 = u$

(4) $c \in F$, $u, v \in V \Rightarrow c*(u+v) = c*u + c*v$

(5) $c_1, c_2 \in F$, $u \in V \Rightarrow (c_1*c_2)*u = c_1*(c_2*u)$

Def:

A non empty set V , equipped with two binary operations i.e. vector addition & scalar multiplication is said to be vector space over a field F , provided V is an abelian group (a-e) w.r.t vector addition and w.r.t to scalar multiplication it satisfies (1-5).

w.r.t
vector
addition
*

Abelian
group

w.r.t
scalar
multiplication
-

Eg: ① The set of all 2×3 matrices is a vector space. with matrix addition and matrix scalar multiplication.

② set of all 3×1 matrices form a vector space.

$$\text{let } u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}; \quad v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}, \quad a, b \in F \text{ scalars}$$

$$w = au + bv = \begin{pmatrix} au_1 + bv_1 \\ au_2 + bv_2 \\ au_3 + bv_3 \end{pmatrix}$$

③ $\mathbb{R}^n, \mathbb{C}^n, P_n(\mathbb{R}); M_{m \times n}$

3×1 matrix $\rightarrow$ closure.

Is Boolean system form a vector space?

Subspace: A subset W of a vector space V is called a subspace of V if W is itself a vector space with same binary operation of V .

Eg: ① The set W of all matrices of the form $\begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ is a subset of (subspace of) $M_{2 \times 2}$.

② W of all matrices of the form $\begin{bmatrix} a & a+1 \\ 0 & b \end{bmatrix}$ is W a subspace of $M_{2 \times 2}$?

Soln Not a subspace of $M_{2 \times 2}$

It satisfies closure & associative but

additive identity $\Rightarrow$ zero matrix does not exist

$\Rightarrow W$ is not a subspace of $M_{2 \times 2}$.

③ Lines through origin are subspaces of $\mathbb{R}^2$ & $\mathbb{R}^3$

Spanning set: If $S = \{v_1, v_2, \dots, v_k\}$ is a set of vectors in a vector space V , then set of all the linear combination of $v_1, v_2, \dots, v_k$ is called span.

$$\Rightarrow u = c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

Q: Set of even numbers, form a vector space? Ans NO!

Q: " odd numbers form a vector space? Ans NO!

Q: Set of Colours, force, Economic data, protein structure. Ans yes.

$\rightarrow$ monochromatic (subspace)

$\downarrow$ Economic indicator (subspace)

matrix < Symmetric matrices
Diagonal matrices

Q. Show that $M_{23} = \text{Span}(E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23})$, where E_{ij} is the matrix with 1 at row i , column j and 0 elsewhere. $E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Soln

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = a_{11}E_{11} + a_{12}E_{12} + a_{13}E_{13} + a_{21}E_{21} + a_{22}E_{22} + a_{23}E_{23}$$

Linearly independent:

The set of vectors $\{u_1, u_2, \dots, u_n\}$ are linearly independent if $c_1u_1 + c_2u_2 + \dots + c_nu_n = 0$ has the only solution.

$$c_1 = c_2 = \dots = c_n = 0.$$

$\Rightarrow$ No vector in set can be written as a linear combination of other vectors.

eg: ① $u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $v = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$; $w = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ are linearly independent?

clearly $w = 2u + 3v \rightarrow w$ is linear combination of u & v

$\Rightarrow$ linearly dependent.

$$\textcircled{2} \quad u = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, v = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, w = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$au + bv + cw = 0$$

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow a = b = c = 0$$

linearly independent

③ which of the following set of vectors are linearly independent?

$$\textcircled{a} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}, \quad \textcircled{b} = \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

$$\textcircled{c} = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

Soln $a, b \rightarrow$ independent, $c \rightarrow$ dependent.

* consider set $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}$

span the vector space of all 3×1 matrices with zero in 3^{rd} row.

$\Rightarrow$ clearly one of the vectors linearly depend on other two.

Basis $\rightarrow$ The set of minimum number of vectors that span the vector space.

$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\}, \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \right\} \Rightarrow$ three combinations from a basis
Special orthonormal basis.

* Let $A = \{u_1, u_2, \dots, u_n\}$ be a basis for a vector space V

a. Any set more than n vectors in V must be linearly dependent

b. Any set of fewer than n vectors in V can not span V .

Dimension: The number of ~~basis~~ vectors in a basis gives the dimension of the vector space.

Eg: previous example $\dim = 2$.

prob:- $M_{2 \times 2}$ is set of all 2×2 matrices, and

$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \right\}$; Check S is a basis of $M_{2 \times 2}$

soln $\Rightarrow \dim(M_{2 \times 2}) = 4$.

$S \rightarrow 3$ elements.

$\therefore S$ can not span $M_{2 \times 2}$.

$\therefore S$ is not a basis of $M_{2 \times 2}$.

matrix spaces

(3)

for $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Then the subspace of $\mathbb{R}^n$ spanned by the row vectors of A is denoted by $\text{row}(A)$ and is called the row space of A , and the subspace of $\mathbb{R}^m$ spanned by the column vectors of A is denoted by $\text{col}(A)$ and is called as column space of A .

The solution space of the homogeneous system of equations $AX=0$, which is a subspace of $\mathbb{R}^n$, is denoted by $\text{null}(A)$ and is called the null space of A .

Column space

eg: $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{pmatrix}$

$$= x_1 \begin{pmatrix} a \\ c \end{pmatrix} + x_2 \begin{pmatrix} b \\ d \end{pmatrix}$$

= linear combination of columns of A .

Q. find the basis for $\text{col}(A)$ and dimension.

for $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & 4 & 5 & 8 & -4 \end{bmatrix}$

Soln) to get the basis, first need to reduce row echelon form. need

$$\text{rref}(A) = \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

↑ ↑
pivot columns

$$\Rightarrow C_2 \rightarrow 2 \times C_1$$

i.e $C_1, C_3 \rightarrow$ independent columns

other columns depends on C_1, C_2

$$\therefore \text{col}(A) = \text{span} \left\{ \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 6 \\ -2 \\ 4 \end{pmatrix} \right\} ; \dim(\text{col}(A)) = 2$$

Null space

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

for Nullspace $AX=0$

$$\rightarrow \text{rref}(A) = \begin{bmatrix} 1 & -2 & 6 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} \rightarrow x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \end{aligned}$$

based on pivot columns $\Rightarrow x_1, x_3 \rightarrow$ basic variables.
 $x_2, x_4, x_5 \rightarrow$ free variables.

Take any value.

$$\begin{aligned} \therefore x_1 &= 2x_2 + x_4 - 3x_5 \\ x_3 &= -2x_4 + 2x_5 \end{aligned} \Rightarrow$$
$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore \text{Basis of Null}(A) = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\text{Null}(A) = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{pmatrix} \right)$$

Prob: Find the basis for null space of $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix} ; \text{rref}(A) = \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$$

$$x_1 = -2x_4$$

$$x_2 = x_4$$

$$x_3 = x_4$$

$$\Rightarrow \begin{bmatrix} -2x_4 \\ x_4 \\ x_4 \\ x_4 \end{bmatrix} = x_4 \begin{bmatrix} -2 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

x_4 free variable

$$\text{Null}(A) = \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

Application of nullspace

(4)

$Ax=b$; fewer equations than unknowns

rows < # columns

→ underdetermined.

let u be general ~~solution~~^{vector} of $\text{Null}(A)$. $\Rightarrow Au=0$

let v be any vector that solves $Ax=b \Rightarrow Av=b$.

then $x=u+v$, be the general soln to $Ax=b$.

$$\begin{aligned}\therefore \Rightarrow Ax &= A(u+v) \\ &= Au + Av \\ &= 0 + b = b.\end{aligned}$$

Eg:- $2x_1 + 2x_2 + x_3 = 0$
 $2x_1 - 2x_2 - x_3 = 1$

$$\Rightarrow \begin{pmatrix} 2 & 2 & 1 \\ 2 & -2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Augmented $\begin{pmatrix} 2 & 2 & 1 & 0 \\ 2 & -2 & -1 & 1 \end{pmatrix}$

part ①

rref $\begin{pmatrix} 1 & 0 & 0 & 1/4 \\ 0 & 1 & 1/2 & -1/4 \end{pmatrix}$

$Au=0 \rightarrow$ forget u^{th} column

$$u_1 = 0$$

$$u_2 + u_3(1/2) = 0$$

u_3 free variable.

$$\text{Null}(A) = \text{span} \left\{ \begin{pmatrix} 0 \\ -1/2 \\ 1 \end{pmatrix} \right\}$$

part ②

particular solution (inhomogeneous) $\Rightarrow Av=b$.

$$x_1 = 1/4$$

$$x_2 + x_3 = -1/4$$

→ NO unique solution

so take free variable as $\rightarrow 0$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = a \begin{pmatrix} 0 \\ -1/2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1/4 \\ -1/4 \\ 0 \end{pmatrix}$$

for any value.

family of solns

Row Space

Find a basis for the row space of $A = \begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$

for row space $\rightarrow$ we take A^T ;

$\Rightarrow$ row space of $A^T \rightarrow \text{col}(A^T)$.

$$A^T = \begin{bmatrix} 1 & 2 & 0 & 2 \\ -2 & -5 & 5 & 6 \\ 0 & -3 & 15 & 18 \\ 0 & -2 & 10 & 8 \\ 3 & 6 & 0 & 6 \end{bmatrix}$$

$$\text{rref}(A^T) = \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & -5 & -10 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\uparrow \quad \uparrow \quad \uparrow$

basis of $\text{col}(A^T) =$ basis of $\text{row}(A)$

$$= \left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ -5 \\ -3 \\ -2 \\ 6 \end{pmatrix}, \begin{pmatrix} 2 \\ 6 \\ 18 \\ 8 \\ 6 \end{pmatrix} \right\}$$

Relation between four subspaces:

A is $n \times n$ matrix, then.

row space: $\text{Row}(A)$: subspace of $(n \times 1)$ matrices.

null space: $\text{Null}(A)$: n

column space: $\text{Col}(A)$: n

left null space: $\text{Null}(A^T)$: n

$(n \times 1)$ n

$(n \times 1)$ n

$(n \times 1)$ n

$\rightarrow \text{col}(A^T)$

$\mathbb{R}^n$

$\mathbb{R}^n$

$\rightarrow \mathbb{R}^m$

$\mathbb{R}^m$

null space: $AX=0 \rightarrow$

• null space is the set of all vectors that are orthogonal to the row space of A .

* $\dim(\text{Null}(A)) = \#$ of non-pivot columns of rref

$\dim(\text{Row}(A)) = \#$ of pivot columns.

Sum of these two subspaces $\rightarrow$ all $(n \times 1)$ matrices
 $\rightarrow \#$ of columns.

$\Rightarrow \text{Null}(A) \perp \text{Row}(A)$ are orthogonal complements of each other.

$$\begin{aligned} \dim(\text{Col}(A)) &= \# \text{ of pivot columns} \\ &= \dim(\text{Row}(A)) \\ &= \text{Rank}(A). \end{aligned}$$

$\Rightarrow \# \text{ of linearly independent rows} = \# \text{ of linearly independent columns.}$

Q2 Find the $\text{Col}(A)$, $\text{row}(A)$, $\text{Null}(A)$, $\text{left}(A)$.

$$A = \begin{bmatrix} 2 & 3 & -1 & 1 & 2 \\ -1 & -1 & 0 & -1 & 1 \\ 1 & 2 & -1 & 1 & 1 \\ 1 & -2 & 3 & -1 & -3 \end{bmatrix} \rightarrow \text{rref} A = \begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & -1 & 1 & 1 \\ 3 & -1 & 2 & -2 \\ -1 & 0 & -1 & 3 \\ 1 & -1 & 1 & -1 \\ 2 & 1 & 1 & -3 \end{bmatrix} \rightarrow \text{rref} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$A \rightarrow \text{pivot columns} \rightarrow C_1, C_2, C_4$

$A^T \rightarrow \text{pivot columns} \rightarrow C_1, C_2, C_3$

$$\therefore \text{Col}(A) = \text{span} \left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \\ -2 \\ -2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ -1 \\ -1 \end{pmatrix} \right\}$$

$$\text{Row}(A) = \text{Col}(A^T) = \text{span} \left\{ \begin{pmatrix} 2 \\ 3 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix} \right\}$$

$$\text{Null}(A) = \begin{aligned} x_1 + x_3 - x_5 &= 0 \\ x_2 - x_3 + 2x_5 &= 0 \\ x_4 - 2x_5 &= 0 \end{aligned}$$

from $\text{rref}(A)$

$$\Rightarrow \begin{aligned} x_4 &= -x_3 + x_5 \\ x_2 &= x_3 - 2x_5 \\ x_4 &= 2x_5 \end{aligned}$$

free variable x_3

$$\Rightarrow x_3 \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 1 \\ -2 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

$$\therefore \text{Null}(A) = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ 2 \\ 1 \end{pmatrix} \right\}$$

$$\text{Null}(A^T) = \begin{cases} x_1 + 2x_4 = 0 \\ x_2 - 2x_4 = 0 \\ x_3 - 5x_4 = 0 \end{cases}$$

from row(A) $\rightarrow$

$$\Rightarrow \begin{cases} x_1 = -2x_4 \\ x_2 = 2x_4 \\ x_3 = 5x_4 \end{cases} \Rightarrow x_4 \begin{bmatrix} -2 \\ 2 \\ 5 \\ 1 \end{bmatrix}$$

x_4 $\rightarrow$ free variable

$$\text{left Null}(A) = \text{span} \left\{ \begin{pmatrix} -2 \\ 2 \\ 5 \\ 1 \end{pmatrix} \right\}$$

check

$$A_{4 \times 5}; \quad \begin{matrix} m=4 \\ n=5 \\ \text{rank}=3 \end{matrix}$$

from row(A) Null(A)

$$\textcircled{1} \text{ Row}(A) \perp \text{Null}(A) \Rightarrow \text{orthogonal.} \rightarrow \begin{pmatrix} 2 \\ 3 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$\textcircled{2} \text{ Left Null}(A), \text{Col}(A) \Rightarrow \text{orthogonal!} \quad = -2 + 3 - 1 = 0$$

$$\textcircled{3} \text{ Rank} = r = \dim(\text{Col}(A)) = 3 \\ = \dim(\text{row}(A)) = 3.$$

$$\textcircled{4} \text{ Nullspace} \Rightarrow \dim(\text{Null}(A)) = 2 = n - r = 5 - 3 = 2$$

$$\textcircled{5} \text{ left nullspace} \Rightarrow \dim(\text{left}(A)) = 1 = m - r = 4 - 3 = 1.$$

Big picture

